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CO1-Assignment Tool 3
Coding Challenge

Course code: DSA0308

Course Name: Natural Language Processing

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1. Intelligent email and password validator using Regular Expressions

```
import re
email = input("Enter Email Address: ")
password = input("Enter Password: ")
mobile = input("Enter Mobile Number: ")
email_pattern = r"^[A-Za-z][A-Za-z0-9._]*@[A-Za-z]+\.(com|org|edu|net|in)$"
password_pattern = r"^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#$%&!]).{8,}$"
mobile_pattern = r"^[6-9]\d{9}$"
if re.match(email_pattern, email):
   print("Valid Email")
else:
   print("Invalid Email")
if re.match(password_pattern, password):
   print("Strong Password")
else:
   print("Weak Password")
if re.match(mobile_pattern, mobile):
   print("Valid Mobile Number")
else:
   print("Invalid Mobile Number")
```

Output:

```
= RESTART: C:/Users/YASWITHA/AppData/Local/Programs/Python/Python314/nlp at3 2.p
y
Enter Email Address: yashu0321@gmail.com
Enter Password: 12345
Enter Mobile Number: 9234167804
Valid Email
Weak Password
Valid Mobile Number
```

2. Design a Finite State Automata (FSA) simulator

```
states = input("Enter states: ").split()
alphabet = input("Enter input alphabet: ").split()
transition = {}
print("Enter transition table:")
for state in states:
   transition[state] = {}
   for symbol in alphabet:
      next_state = input(f"Delta({state}, {symbol}) = ")
      transition[state][symbol] = next_state
initial = input("Enter initial state: ")
final_states = input("Enter final states: ").split()
n = int(input("Enter number of input strings: "))
for i in range(n):
   string = input("Enter input string: ")
```

```
    current = initial
    path = [current]
    valid = True
    for symbol in string:
        if symbol not in alphabet:
            valid = False
            break
        current = transition[current][symbol]
        path.append(current)
    print("Transition Path:")
    print(" -> ".join(path))
    if valid and current in final_states:
        print("Accepted")
    else:
        print("Rejected")
```

Output:

```
= RESTART: C:/Users/YASWITHA/AppData/Local/Programs/Python/Python314/nlp at3 2.p
y
Enter states: q0 q1 q2
Enter input alphabet: a b
Enter transition table:
Delta(q0, a) = q1
Delta(q0, b) = q0
Delta(q1, a) = q1
Delta(q1, b) = q2
Delta(q2, a) = q1
Delta(q2, b) = q0
Enter initial state: q0
Enter final states: q2
Enter number of input strings: 2
Enter input string: abab
Transition Path:
q0 -> q1 -> q2 -> q1 -> q2
Accepted
Enter input string:
```

3. Smart Pattern Matching Engine using Regular Expressions

```
import re
text = """Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing"""
while True:
    print("\nUser Menu")
    print("1. Search Word")
    print("2. Search Date")
    print("3. Search Phone Number")
    print("4. Search Hashtag")
    print("5. Search Mention")
    print("6. Search Prefix")
    print("7. Search Suffix")
    print("8. Exit")
    choice = input("Enter your choice: ")
    if choice == "1":
        word = input("Enter word to search: ")
        pattern = r"\b" + re.escape(word) + r"\b"
    elif choice == "2":
        pattern = r"\b\d{2}/\d{2}/\d{4}\b"
    elif choice == "3":
```

```
    pattern = r"\b[6-9]\d{9}\b"
elif choice == "4":
    pattern = r"#[A-Za-z0-9_]+"
elif choice == "5":
    pattern = r"@[A-Za-z0-9_]+"
elif choice == "6":
    prefix = input("Enter prefix: ")
    pattern = r"\b" + re.escape(prefix) + r"\w*"
elif choice == "7":
    suffix = input("Enter suffix: ")
    pattern = r"\w*" + re.escape(suffix) + r"\b"
elif choice == "8":
    print("Exiting...")
    break
else:
    print("Invalid choice")
    continue
matches = re.findall(pattern, text, re.IGNORECASE)
if matches:
    print("Matching Patterns:")
    for match in matches:
        print(match)
else:
    print("No matching patterns found")
```

Output:

```
= RESTART: C:/Users/YASWITHA/AppData/Local/Programs/Python/Python314/nlp at3 2.p
y

User Menu
1. Search Word
2. Search Date
3. Search Phone Number
4. Search Hashtag
5. Search Mention
6. Search Prefix
7. Search Suffix
8. Exit
Enter your choice: 1
Enter word to search: Clean
No matching patterns found

User Menu
1. Search Word
2. Search Date
3. Search Phone Number
4. Search Hashtag
5. Search Mention
6. Search Prefix
7. Search Suffix
8. Exit
Enter your choice: 2
Matching Patterns:
12/09/2026
```